

Coursework Specification

Read this coursework specification carefully, it tells you how you are going to be assessed, how to submit your coursework on-time and how (and when) you'll receive your marks and feedback.

Module Code	CSI_5_OSY
Module Title	Operating Systems
Lecturer	Ioannis Iatropoulos
% of Module Mark	60%
Distributed	10/10/2025
Submission Method	Submit online via this Module's Moodle site
Submission Deadline	9 th Dec 2025, at 23:30
Release of Feedback & Marks	Feedback and provisional marks will be available in the Gradebook and Turnitin (Rubric) on Moodle within 2-3 weeks from the submission deadline.

Coursework Aim:

The coursework assignment requires from students to make an implementation on a Raspberry-Pi. This is a group coursework, with maximum number of two students per group. The groups will be defined during the first lab of the semester.

On this assessment, the students must choose **ONE** of the suggested projects for implementation on a Raspberry-Pi. They will have to build a working prototype of that application, benchmarking and resource monitoring, evaluate key performance indicator markings, and examine the system scalability. Then submit a scientific report individually, along with any supporting materials, and demonstrating their project in a video presentation.

The project for implementation must be selected from the in-scope list given in the "Assessment Task" section, otherwise may be considered out-of-scope.

This coursework will test the students on their knowledge and understanding of operating systems, managing and monitoring resources, handling processes and threads, allowing them to demonstrate these skills in a practical test case by utilizing a Raspberry-Pi.
(See Assignment Task, below).

Coursework Details:

Type:	Project Report, Group Presentation, Project materials
Word Count:	A minimum of 2500 words is recommended, with 5000 words maximum limit. The following texts will be excluded from the word counts. • Footnotes for reference purposes only. • Bibliography. • List of tables, Figures and Table of Contents. • Source code and scripts. • Everything that is included in the Appendix section. You might append source codes, development environments, and any additional resources at the appendix section. However, these will be excluded from the marking scope. Please note in case you develop your own tool or code; the complete source code must be submitted as a separate script for evaluation and correctness as supporting materials.
Document Structure and Readability:	▪ The document is Recommended to follow the format from one of the suggested Project report templates provided under the Learning and Support section on VLE (https://overleaf.com). ▪ Work must be cited through appropriate bibliography. ▪ Work must be submitted as a Word document (.doc/docx) or a PDF (not exceeding 100MB). ▪ The coursework document must follow a consistent template, using a single font and font size (e.g., Arial 11 or larger if you need to for the headings), with a recommended 1.5 line spacing. ▪ Any code analysis or command execution must be complete and concise with proper commenting to explain the logic, attributes/options/flags and input parameters. To present the Source Code snippets, command execution, bash scripts, or configuration files, you need to use a fixed-width font and clearly readable (no screenshots), recommended Monospaced fonts (Consolas, Courier New, etc.). You might split the code by module and present it through a table (1x1) used as a code-box, that will clearly separate from the narrative. ▪ Your student ID number must appear on the front page of the coursework. Your name must not be on your coursework.

Screenshots Requirements:	▪ Each student after installing the Operating System in the RaspberryPi, they need to create up to two new system users (one for each student in the group) with their username along with their last two digits of their student-ID. The username could be your first name, your surname or a combination from the first letters of both, for example “ioaniatr80”. (Example at the end of this document) ▪ The screenshots need to include your user on each command execution, unless the command requires the ‘root / administrator’ user and cannot be executed with “sudo / run as administrator”. (Example at the end of this document) ▪ Screenshots must have a captioned, cross-referenced and should be used mostly to provide evidence of: ○ Output or results of a program/tool execution (any command that has been executed as an input must be provided also by text). ○ Designs, such as Information on the network environment, IP addresses, activity diagrams/flowcharts etc. ○ Present a software interface (GUI/CLI), or tools. ○ Graphical Analysis may include a representation of the results or similar type of comparative plots (e.g., for resource monitoring or performance evaluation). ○ Any other relevant evidence that can be used as proof of concept. Note: Screenshots which do not satisfy the above-mentioned requirements will not be considered and the report will be penalised accordingly!
Referencing:	Harvard Referencing should be used, see your Library Subject Guide for guides and tips on referencing. The references must include at least two (or more) of following types of sources: (Mandatory) A reference with the link of the selected project and its origin, along with the name of the application/software (if applicable). (Mandatory/Optional) References of any installation tutorials or usage guides that have been followed. (Optional) References of any forums, or links that have

	been followed, along with access date and title. (Optional) Books and other resources studied to accomplish these tasks. Note: Inappropriate reference formats or mixed referencing will be penalised as stated in the assessment rubric's report structure and readability criteria!
The Content:	• The final report must document all the steps, commands issued, and console output in the form of a scientific format. • The documentation should be thorough enough that the implementation can be replicated step-by-step by a technically incompetent reader. Note: Your report should be narrative in style, with human explanation and commentary. A "report" that is merely a collection of screenshots and data dumps will be graded very poorly.
Video Presentation:	The video presentation must include a summary of your work in the form of a short video (.mp4), duration up to 10 minutes max, explaining your work, and a sample demo is required.
Use of Generative AI:	In this module, students are permitted to use generative AI tools, such as ChatGPT or Copilot, to assist with writing code and technical reports. However, students must include the following in their technical report: • Acknowledgment of AI use. • Details on the extent of use, and how the generated content was applied. • Descriptions of the prompts and how AI-generated content was applied. While AI can offer useful guidance, it tends to make errors and may not fully align with the coursework requirements, potentially affecting marks. Students are encouraged to understand the assignment thoroughly and work independently. Starting early, following lab sessions closely, and completing the work alongside workshop content are crucial for success. AI can be used for suggestions and refinement, but students must provide evidence of practical implementation, especially in tasks like Raspberry Pi hardware configurations and measurements, which AI cannot perform.

Submission:	Students could submit up to 3 different FILES: i. (Mandatory) The main report (.doc, .docx, .pdf), which includes the documentation of the coursework in a scientific format. ii. (Optional) A compressed file (.zip), with the supporting materials e.g., source code or scripts that has been developed (.sh, .py, .cpp, .java), tool or release versions (executable file .pyc, .exe, .out, .jar etc.) that are not included into the default Kali-Linux installation, references/links with code snippets that has been used, dependencies (e.g., libraries/directives), or other kind of software that is required to recreate the testing environment etc. In case the file size exceeds 100MB, a cloud storage should be used (OneDrive, Google drive, etc.) with appropriate access rights, and submit a single text file (.txt) that will include the link. This is optional, as it is supposed that those materials should be included in the SD-Card that needs to be delivered to your tutor at the end of the project. Supporting materials are required to satisfy some of the criteria as implementation evidence. iii. (Mandatory) The audiovisual presentation (.mp4), summarizing the work. Alternatively, in the case of a large video file, share it via a cloud solution (e.g., oneDrive, Youtube etc.). In this case, the link should be submitted through a simple text file (e.g., "presentation.txt"). Note: Any submission with inaccessible material(s) will not be considered for marking. The filenames need to have the format: i. studentID_name-coursework2_report.doc (example: 123456_John_Doe-coursework2_report.doc) ii. studentID_name-project_materials.zip (example: 123456_John_Doe-project_materials.zip) iii. studentID_name-presentation.mp4 (example: 123456_John_Doe-presentation.mp4) ➔ Optionally, for bonus marks, students will have the opportunity to receive feedback on their progress and demonstrate their implementation plan by submitting a
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	draft through a dedicated link on the VLE midway of the semester.
Regulations:	Make sure you understand the University Regulations on expected academic practice and academic misconduct. Note in particular: ▪ Your work must be your own. Markers will be attentive to both the plausibility of the sources provided as well as the consistency and approach to the writing of the work. Simply, if you do the research and reading, and then write it up on your own, giving reference to sources, you will approach the work in the appropriate way and will not give markers reason to question the authenticity of the work. ▪ All quotations must be credited and properly referenced. Paraphrasing is still regarded as plagiarism if you fail to acknowledge the source for the ideas being expressed. TURNITIN: When you upload your work to the Moodle site, it will be checked by anti-plagiarism software. Your similarity index for the report must not be more than 20%. Any report with exceeding a 20% similarity index will be subject to Academic Misconduct Investigation (AMI).

Learning Outcomes

By completing this coursework, students will be able to:

Hardware and Embedded Systems

- Identify and describe the key hardware components of a Raspberry Pi, including GPIO pins, interfaces, and peripheral connections.
- Demonstrate an understanding of how embedded systems operate within a hardware environment and interact with connected devices.
- Explain the functionality and configuration of operating systems designed for embedded platforms such as the Raspberry Pi.

System Configuration and Networking

- Set up, configure, and manage network connections on a Raspberry Pi, applying appropriate networking concepts, protocols, and security principles.
- Analyse and evaluate operating system mechanisms such as process management, memory allocation, and resource handling for various applications.

- Implement software and hardware configurations that demonstrate the integration of operating system components with hardware resources.

Design, Implementation, and Innovation

- Design and implement an innovative embedded system application using the Raspberry Pi, integrating creativity, usability, and technical effectiveness.
- Apply user-centered design principles to develop intuitive and efficient interfaces for Raspberry Pi-based applications.
- Develop, test, and refine software solutions through experimentation, benchmarking, and performance evaluation under varying conditions.

Problem-Solving and Evaluation

- Diagnose and resolve hardware and software issues encountered during development, applying systematic troubleshooting strategies.
- Conduct resource monitoring and performance analysis to evaluate application scalability, efficiency, and reliability.
- Interpret experimental results and provide critical analysis supported by data visualization and benchmarking metrics.

Project Management and Professional Practice

- Plan and manage the project effectively by defining objectives, organizing tasks, and meeting deadlines.
- Demonstrate responsibility in handling, maintaining, and returning laboratory equipment in accordance with institutional requirements.
- Collaborate effectively within a team to achieve shared goals, demonstrating clear communication, accountability, and contribution to group outcomes.
- Produce a well-documented report and video presentation that clearly communicate the project design, implementation, and evaluation results.

Knowledge Integration

- Synthesize knowledge of operating systems, embedded systems, and networking to develop comprehensive, real-world solutions.
- Demonstrate a holistic understanding of the relationship between software, hardware, and system-level performance in embedded computing environments.

Assessment Tasks

This coursework is group based and weighs **60%** of the overall marks.

The primary objective of this assessment is to perform a software implementation on a Raspberry-Pi, monitor resources, understand how the operating system is managing those resources, and make a performance evaluation. It is also expected to demonstrate their application in a short video duration 10 minutes maximum. For this project, students will have to utilize specialized equipment that will be available only on-campus or during the labs.

Responsibility and Care of Equipment:

Once students have finalised their team formation and chosen their project topic, they will be provided with an equipment box that will include a Raspberry-Pi along with all the necessary required to completed this assessment. Following professional standards, students are responsible for maintaining the lab equipment in good condition throughout the assessment. A checklist of all components in the equipment boxes should be kept to ensure accountability. To provide students the equipment box, and allow them to take it off-campus they must first complete the following in this particular order:

1. Create a team and choose their Topic by submitting online the "[Team Allocation Form](#)".
2. Submit the "[Equipment_Allocation_Form](#)" document on VLE.

All equipment must be returned intact and undamaged by the next day of the submission deadline. Further instructions for the "*Coursework Roadmap*" until your final submission and how to complete those steps are provide on VLE under the "[Assessments and Feedback](#)" section.

Note: Any damaged or lost equipment must be replaced by the student responsible. **Failure to do so will reflect in their module grade** under the the "*Responsibility and Care of Equipment*" criteria, with grade deductions applied for late returns, damage, or poor condition.

Teamwork:

During this coursework, students will form teams consisting of two members working in pairs to complete the assigned tasks. Each team must independently develop its own project and should not engage in the development of another team's task, even if the teams share the same topic.

Teams should ensure that their efforts do not hinder the development of other teams' projects otherwise this may lead to AMI. If there are any concerns about the potential exposure of sensitive aspects of your project, it is recommended to consult with the module leader or tutors beforehand to safeguard your work and ensure a smooth collaboration process.

As part of this collaboration, each student is required to submit an **individual report** and a **video presentation**. The report should be based on their own experiences, demonstrating their understanding of the group's achievements. The report must outline roles, responsibilities, communication dynamics, and the strengths or weaknesses observed within the team.

All reports will be subjected to similarity analysis to identify students who have not written in their own words but used the same writing as others.

Video Presentation:

The video will be evaluated based on the following criteria:

- Presentation format / consistency
- Duration
- Coverage and scope
- Fluency and clarity
- Domain of knowledge
- Demonstration / Soft simulation
- Teamwork

To minimize the risk of malpractice, **you must appear** within the majority of the recorded evidence. If you are including a 10-minute presentation video as part of your final submission, then you must **show your student ID** at the beginning of the video to verify that it is you.

The video presentation would contribute up to the **maximum 10 minutes** of submission allowable, and should not be exceeded. If the duration of 10 minutes is reached, the remaining evidence will not be considered, and anything over this limit will be disregarded.

Draft Submission:

Students may optionally submit a draft of their coursework by the **deadline of 11/11/2024** via a dedicated link on VLE to receive formative feedback. The draft will be assessed for consistent effort on the project up to the draft's submission date, and thoughtful planning for achieving final goals. Feedback on the draft will be provided during the evaluation of your draft submission (within 15 days), but the bonus marks will be awarded accordingly to the marking criteria in the rubric of the final coursework submission. The primary aim of this submission is to provide feedback to help you improve your work for the final submission.

Draft submissions should demonstrate the development of a computer system or formal specification with appropriate complexity and originality, including

explanations of code, testing, methodology, and problem-solving. Students should also provide a realistic plan outlining their approach, practical activities, and report writing, ensuring a clear roadmap for successful project completion while rewarding ongoing effort and careful planning.

The coursework assessment is mainly divided into two tasks as follows.

Task 1 – Application Implementation

Domain of Knowledge:

Students must show in-depth understanding of the application, describing the application and its features. Moreover, must include a discussion of the ways in which the program relates to the theoretical concepts covered in the module, by presenting this application from the aspect of the operating system. In particular, explain how the operating system handles this application, interacting with hardware/software resources, processes and threads that are spawned, users involved, access level (user/kernel), locations in the file system used, examine any concurrency issues, present requirements, design system model and application workflow e.t.c.

Implementation:

Be Aware: All projects **must be completed using actual Raspberry Pi hardware** and in collaboration with your team. The implementation of the coursework ***should NOT*** be made on a virtual machine (e.g., VirtualBox or another hypervisor), as it may **negatively affect your mark** and could be considered off-topic. The primary objective of this assessment is to **utilize and experiment with the OS on Raspberry Pi as physical hardware**.

The students must choose **ONE** of the following “In-Scope” projects list for implementation on a Raspberry-Pi, and configure making an effective use of operating system facilities:

1. Baby Monitor:

- In this implementation, students are required to design and implement a Baby Monitor system using a Raspberry Pi platform. The system should be capable of **detecting when a baby is crying** and, upon detection, automatically sending notifications or alerts (e.g., via email) to a remote device such as a smartphone or computer. In addition, the monitor should be able **to remotely play soothing sounds (“lullabies”)** from a selection of white noise and lullabies to help the baby fall asleep when necessary. Students must install an appropriate operating system (such as Raspberry

Pi OS) on the Raspberry Pi, which will host a service or framework responsible for **CCTV surveillance**.

- The proposed solution must support and integrate the Raspberry Pi camera module for capturing high-quality video and providing real-time streaming capabilities. Optionally, students may include additional features such as adjustable video settings (e.g., resolution, bitrate, and frame rate) to enhance performance. Furthermore, each implementation must utilise a USB sound card and an external microphone (will be provided) for audio capture and cry detection . For cry detection you could use the “[Baby Crying Sound Effect samples](#)” provided on VLE. A user-friendly graphical (GUI) or command-line (CLI) interface should be developed for managing the streaming service (e.g., viewing the camera feed) and control system functionalities (e.g., playing lullabies), ensuring compatibility across various devices for seamless access and control. The final system should demonstrate effective integration of audio detection, camera functionality, and network communication, operating reliably, efficiently, and in real time.

2. Smart TV:

- In this task, students are required to design and implement a Smart TV system using a Raspberry Pi, capable of **local media playback** and **online streaming** through a remote-friendly interface. They will install and configure of at least two operating system solutions (e.g., OSMC and LibreELEC) and compare the basic features each provides, including media apps, navigation, and playback performance. Following this comparison, students should select and **justify one preferred solution for further development**, adding enhancements such as modular app channels (YouTube, Local Media, Live/IPTV), voice search, screensaver functionality, remote control via phone with DPAD navigation, and a developer-oriented performance overlay with a web dashboard to monitor system metrics and export performance data.
- This project provides hands-on experience in embedded systems, multimedia handling, web development, and performance benchmarking, requiring students to evaluate trade-offs and implement additional functionalities on a single platform.

3. MS-Dos:

- In this task, students are required to design and implement a **fully functional MS-DOS environment** on a Raspberry Pi using an MS-DOS emulator such as DOSBox. The emulator must be installed on a Debian-based Linux distribution on the Raspberry Pi and configured to provide a complete DOS terminal experience. This should include command-line utilities (e.g., MEM.EXE, CHKDSK.EXE, FDISK.EXE, UNDELETE.EXE,

MSD.EXE), file management tools (e.g., DOS Navigator, XTreeGold), text editors (e.g., EDIT.COM), and other MS-DOS applications (e.g., DOSKEY.COM, QBASIC, Turbo C, Turbo Pascal). Students are expected to expand the limited functionalities of DOSBox by integrating essential DOS tools and fun utilities such as ASCII demos, CGA/VGA painting programs, and benchmarking tools.

- Students should demonstrate how applications interact with the operating system, manage files, access hardware resources, and handle processes. The report should cover MS-DOS features, workflow, concurrency, user access, and design decisions made to expand the emulator's capabilities. This project provides hands-on experience in legacy OS design, emulator configuration, system programming, and software integration, challenging students to balance authenticity, usability, and technical creativity on a modern embedded platform.
- For the DOS environment, students should consider any of the following options:
 - FreeDOS (DOS Base & Full ISO)
 - Vetusware (Legacy DOS software archive)
 - WinWorldPC (Archives of DOS software and tools)

4. Firewall:

- In this project, students are required to install and configure a **firewall application** (e.g., Netfilter, UFW, Shorewall, pfSense) on a Raspberry Pi, transforming it into a fully functional network security device. The objective is to evaluate students' understanding of networking protocols, including traffic control, packet filtering, routing, and monitoring. The Raspberry Pi will operate as a firewall and gateway between multiple network segments, enforcing security rules to protect and manage network communications while demonstrating controlled and traceable traffic flow.
- This topic requires students to apply practical networking and security skills by configuring and testing firewall rules for traffic management, monitoring, and routing. They must present a clear **network topology** showing the Raspberry Pi as the central firewall, along with a **working demonstration** of its filtering and routing behaviour, along with a **concise report** explaining the firewall's logic, rule design, traffic flow, and security mechanisms.
- The configuration must utilize at least three network interfaces (wired or wireless), with the following setup:
 - One local interface connected to a **Class-C** network, accepting incoming traffic from an access point operating at 2G frequencies.

- One local interface connected to a **Class-B** network, accepting incoming traffic from an access point operating at 5G frequencies.
- One external interface connected to a **Class-A** network, providing internet connectivity for both access points.
- Students are not required to implement physical access points; instead, they may emulate network traffic using any suitable device within their testing environment. The main objective is to design, configure, and secure a robust firewall solution on the Raspberry Pi.
- More specifically, the firewall rules must meet the following key requirements:
 - Default policy should deny all traffic unless explicitly allowed by a defined rule.
 - Allow only essential services and protocols (e.g., DNS, DHCP, ICMP) between networks.
 - Enable packet forwarding to share internet access with internal networks.
 - Permit internal communication between the 2G and 5G access points.
 - Allow traffic forwarding from local networks to the external interface for internet access.
 - Permit return traffic from external networks only for established connections.
 - Deny externally initiated connections to internal hosts.
 - Implement anti-IP spoofing measures.
 - Permit SSH and RDP access from local interfaces only, while denying VNC traffic.
 - Restrict remote administration of the firewall, and allow only authorized hosts (based on static IP or MAC address mapping).
 - Log all dropped or denied packets for analysis and reporting.
 - Any additional, well-justified rules that improve security or functionality will be considered as evidence of comprehensive and advanced work.

Note: In case that students want to use a different implementation that is not included in the “In-scope” list above, please be advised by the module leader, as it may lead out of topic and result with zero marks.

The number of projects for implementation is limited and students must express their interest within the first two weeks of the semester by submitting the “Team Allocation Form”. Otherwise, the projects and teams will be allocated randomly, and be available on VLE at the end of week-3. While every effort will be made to accommodate your preferences, please note that final team and project allocations may differ from your desired choices depending on availability and overall

distribution requirements. All the form submissions will be handled in first-come-first-served basis.

Be advised, once the “Team Allocation Form” submitted, no further changes can be made, and additional responses will not be accepted. Once groups have been defined and projects allocated, you can only work on the allocated project. Changes will not be accepted, and if a different project will be submitted that the one that has been allocated, there is a high risk to be considered out of topic.

Task 2 – Evaluation of the application

The software from the previous task must be examined thoroughly based on use of resources and evaluate its performance and scalability under different circumstances. More specifically:

- **Resource Monitoring:** Based on the nature of the selected application, students must perform benchmarking over a period of time, identify the key performance indicator markings (CPU Utilization, GPU Utilization, thermal conditions, network throughput, and playback stats e.t.c.) for this application, and create an analysis and visualizations of resource monitoring. Students may need to synthesize various resources or tools to accomplish this task.
- **Performance evaluation:** Evaluate the effectiveness of the application based on its role and nature. Provide means to measure the performance of the system, evaluate and assess the system in use by conducting a stress test. For this purpose, various metrics must be tested on cases/profiles that will be created, and present data visually using graphs and charts (in a dashboard or via exported reports). Create different scenarios, for example, under heavy workload, busy network, multiple users, multiple components etc. and finally, interpretation of the results and provide a critically evaluative analysis discussing the scalability of the system.

Assessment Criteria and Weighting

The report will be marked using the marking criteria provided in this coursework specification document. All the students are advised to fully understand the marking criteria before starting the coursework.

LSBU marking criteria have been developed to help tutors give you clear and helpful feedback on your work. They will be applied to your work to help you understand what you have accomplished, how any mark given was arrived at, and how you can improve your work in future.

Marking Criteria

Criteria		Feedforward comments				
	Weighting	Excellent (100-71%)	Good (70-61%)	Pass (60-51%)	Weak (50-41%)	Poor (40-0%)
Report and Documentation						
Report Structure, Readability, and Referencing (Marks 5%)	5%	Sophisticated, consistent, error free application of relevant topics conventions with great attention to detail. Excellent writing, structure, spelling, grammar. The length of the report is more than 2500 words. The references are complete and in the correct format. Used correctly inside the text. Provide valid and reliable information.	Comprehensive application of relevant topics conventions with few errors. Very good writing, structure, spelling, grammar, but with minor errors. The length of the report is more than 2500 words The references are valid and complete, but there are some errors in the format.	Generally correct application of relevant topics conventions, with some errors and / or inconsistencies. The length of the report is at least 2500 words. Sufficiently written with little structure, spelling, grammar with some errors. All sources are valid and reliable, but the references are incomplete or partially used inside the text.	Poorly written with confusing structure, spelling, grammar and / or errors. Below the minimum of 2500 words. The references are incomplete, or no references used inside the text. They include some sources that are not valid or reliable.	Poorly written, less than 2500 words, with no academic style, structure, spelling, grammar and/or multiple errors. The final report does not include any references. Most of the references are not valid.
Abstract and Introduction (Marks 5%)	5%	A well-articulated abstract and introduction that provides a clear, logical, and succinct description of content, objectives, scope and requirements. The organization of the review, which draws the reader's attention to a central concern, debate, or contention.	A well-articulated abstract and introduction that provides a clear, logical description of content, objectives, scope, requirements and organization of the review	Satisfactory abstract and introduction that has a good reflection and description of the content, objective, scope, and organization of the report.	An abstract that articulates some key components of the report. An introduction that outlines the content, scope, and organization of the report	Either no abstract or introduction, or one that poorly or partially situates the reader in the context of the concern, debate, or contention addressed in the report
Conclusion and Recommendations (Marks 5%)	5%	Excellent breadth, accuracy, and detail in understanding key aspects of subject and reflection of the project. Contributes to subject debate. Very good awareness of ambiguities and limitations of knowledge. Provides high-level summary and recommendations are very accurate and detailed.	Good depth of understanding of key aspects of subject, and reflection of the project is shown Evidence of coherent knowledge. Very good contribution to subject debate. Very good summary and accurate recommendations	Demonstrated good understanding of key aspects of subject, with a good reflection of the project. Some evidence of coherent knowledge and own critique. Sufficiently summarize the report with some recommendations	Weak evidence of superficial understanding of subject, with Inaccuracies. Some outline of recommendations, does not summarize well the report or lack of succinctness. Not good reflection of the project.	Little or no evidence of understanding of subject. Inaccuracies. Without any recommendation, does not summarize the report.
Application Implementation						
Domain of Knowledge (Marks 10%)	10%	Shows breadth, accuracy and detail in understanding key aspects of subject. Contributes to subject debate. Some awareness of ambiguities and limitations of knowledge. Knowledge and understanding are consistent and accurately developed with a level of criticality. Exceptional error-free presentation of the software/hardware, and system requirements, with extensive appropriate use of notation. System model and design diagrams provide a comprehensive and insightful set of views of the systems use.	Accurate and extensive understanding of key aspects of subject. Evidence of coherent knowledge. Knowledge and understanding are basic/relatively superficial. A complete and appropriate use of notation to present software/hardware, and system requirements. The System model and design diagrams accurately represent almost all the usage requirements of the system in use.	Accurate understanding of key aspects of subject. Evidence of coherent knowledge. Knowledge and understanding are detailed and satisfactory. Simple design diagrams with reasonable use of notation with some errors. The system model accurately represents the basic use of the system.	Some evidence of superficial understanding of subject. Minor Inaccuracies. Knowledge and understanding shows consistent gaps. Limited use of notation with obvious errors and mistakes. The system model and design diagrams provide a poor or misleading interpretation of the use of the system.	Little or no evidence of understanding of subject. Many Inaccuracies. Knowledge and understanding are poor and lacks academic rigour. Hardly any information provided about the system model.

	Implementation (Marks 20%)	20%	Excellent problem-solving ability and implementation of the proposed methodologies and solutions. Ability to Adapt to unforeseen practical and theoretical challenges to achieve project objectives. Well-crafted technical solution, addressing all aspects of the user requirements.	Very good problem-solving ability and implementation of the proposed methodologies and Solutions. Adapt to practical and theoretical challenges to achieve project objectives. Comprehensive technical solution, addressing various aspects of the user requirements.	Sufficient problem-solving ability and implementation of the proposed methodologies and solutions. Some adaptation to practical and theoretical challenges to achieve project objectives identified goals. Good technical solution, addressing most aspects of the user requirements.	Limited problem-solving ability and implementation of the proposed methodologies and Solutions. Limited exploration of possible solution(s) using established approaches to resolve practical and theoretical problems. Weak attempt at the technical solution, addressing only few aspects of the user requirements.	Poor or lack of problem-solving ability and implementation of the proposed methodologies and solutions. Little or no exploration of solution(s). Question or problem unresolved. Poor attempt at technical proposition.
	Resource Monitoring (Marks 10%)	10%	Very high-quality analysis of the system developed independently. Sustained benchmarking, resource monitoring and synthesis of tools/practices. Use of evidence-based arguments. Thoroughly identifies and explains the key indicator markings. States the system status with extensive level of detail for the handling of the related processes, and other components.	Very good system analysis and use of tools/practices for benchmarking and monitoring of resources. Use of evidence-based arguments. Well identifies most of the key indicator markings, and states the system status on handling processes, data or other components.	Applies practical and/or technical skills correctly and with some application for resource monitoring. Adapts to unforeseen practical and theoretical challenges to achieve identified possible key indicator markings, and sufficiently present the system status.	Some evidence of limited problem-solving ability. Appropriate response to monitor resources, but there is little development and exploration evident of the possible solution(s), and/or weak interpretation of the key indicator markings. Some attempt at evaluation and synthesis of resources. Some use of evidence-based arguments.	Little or no exploration of possible solution(s), no exploration of the available resources. Very little use of evidence-based arguments to present the status of the system.
	Performance Evaluation and Scalability (Marks 10%)	10%	An exceptional level of embodied awareness revealed in a highly sensitive and detailed performance evaluation. Examines all the aspects for scalability of the system.	Well-developed embodied awareness; for the most part. A detailed and accurate responsiveness; Very good use of tools/practices for a dynamic performance evaluation and interpretation of the results. Examines most of the aspects for scalability of the system.	Applies practical and/or technical skills correctly and with a proper use of tools and applications for performance evaluation, that may include minor inaccuracies in the interpretation of the results. Satisfactory presentation of the system's scalability.	Some evidence of limited problem-solving ability. Appropriate response to evaluate performance, but there is little development and exploration evident of the possible solution(s). Weak interpretation of the results, performance evaluation, or scalability of the system.	Little or no exploration of possible solution(s) for performance evaluation, or scalability of the system.
	Project Management: Responsibility and Care of Equipment (Marks 15%)	15%	Students consistently demonstrate exceptional responsibility in handling and returning all equipment, including SD cards, cables, and other components. All items are maintained in excellent condition, with no reports of damage, loss, or late returns. There is clear evidence of proactive care, including proper storage, regular checks, and strict adherence to usage protocols. Collaborative efforts to maintain all equipment are exemplary.	Students generally show good responsibility in managing equipment, including SD cards, cables, and other components. Most items are returned in good condition, with only minor issues noted, and late returns are infrequent. Students follow protocols for care and handling, demonstrating an understanding of proper procedures. There is evidence of collaborative efforts to ensure all equipment is well-maintained.	Students show acceptable responsibility for equipment, including SD cards and cables, with some minor issues regarding care, handling, and timely returns. Some items may show signs of wear or improper handling, and a few late returns may be noted, but no significant damage or loss is reported. Basic adherence to protocols is observed, though further improvement is needed in collaborative maintenance efforts.	Students demonstrate limited responsibility for equipment care, with noticeable issues in handling, returning items late, and adhering to protocols for SD cards, cables, and other components. Some equipment is returned damaged or late, and there are frequent lapses in adherence to care protocols. Evidence of collaborative maintenance efforts is minimal, indicating a need for improvement.	Students show little to no responsibility for equipment, including SD cards and cables, with multiple instances of damage, loss, or late returns reported. Items are consistently returned in poor condition or not returned at all, and there is no adherence to care protocols. There is a lack of evidence demonstrating any collaborative maintenance efforts, indicating a disregard for shared resources.

	Teamwork (Marks 5%)	5%	Team structure, roles and responsibilities are exceptionally well defined. Detailed and fairly distributed, considering team strengths and supported by well executed project management skills. Excellent evidence of active team working and the use of project management tools and techniques.	Team structure, roles and responsibilities are well defined, detailed and fairly distributed, considering most team strengths. Comprehensive evidence of planning, and project management method.	Team structure, roles and responsibilities are defined to some extent, though there is lack of detail. Competent use of project management tools and techniques.	Team structure, roles and responsibilities are poorly defined and/or distributed. Tasks distributed with some unfairness and/or only considering some team strength. Little use of project management tools and techniques.	Little or no evidence of collaboration with others; unproductive working alone. Poor or no evidence of the use of project management tools and techniques.
	Draft Submission (Bonus / Optional) (Marks 10%)	10%	Demonstrates exceptional and consistent effort on project development, with clear, detailed, and realistic planning for final goals. Evidence of sophisticated understanding of the system or specification developed so far. All documentation is comprehensive, well-structured, and clearly communicates progress and next steps.	Shows strong effort and well-thought-out planning. Work completed to date is mostly accurate and coherent, with minor gaps or omissions. Documentation is clear and organized.	Demonstrates satisfactory effort and adequate planning. Work completed shows some progress toward project goals but may lack detail or clarity in areas. Documentation and explanations are present but may be limited or inconsistent.	Shows limited effort or inconsistent progress. Planning is incomplete or lacks realism. Documentation and explanations are insufficient or unclear.	Minimal or insufficient effort on project work. Planning is absent, unrealistic, or incoherent. Documentation is incomplete or poorly presented.
	Presentation						
	Video Presentation (Marks 15%)	15%	Showed excellent confidence & composure. Very clear, persuasive and compelling with skilful use of the presentation format. Presentation addresses the needs of the audience very well. Excellent demonstration of the system, with consistence understanding for the domain of knowledge, and teamwork.	Were mostly confident & composed. presentation is clear, mostly persuasive, compelling and skilfully presented. Presentation addresses demonstration and the needs of the audience to a large degree.	Good use of the presentation format and skills. Presentation considers the needs of the audience, with basic demonstration of the system, and satisfactory information regarding teamwork.	Presentation format is adequate. Showed some confidence & composure but has room for improvement. Presentation may sometimes not consider the needs of the audience. Weak demonstration of the system, and teamwork is not well presented.	Showed no confidence or composure. presentation format is not used adequately, and the needs of the audience are not considered. Poor evidence, or no demonstration of the system. Little or no information about teamwork in the presentation.

How to get help

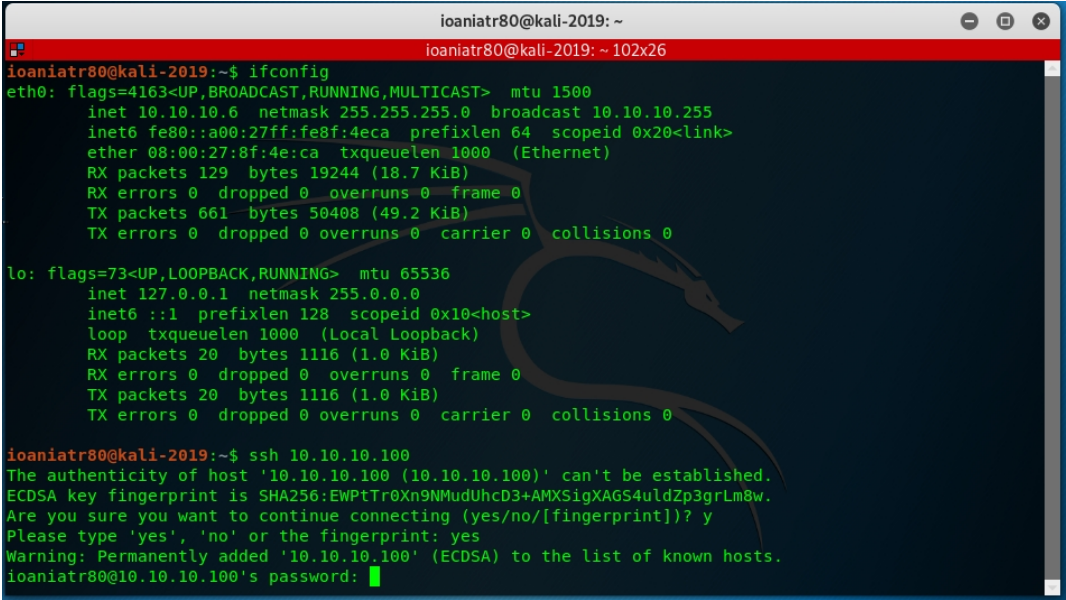
We will discuss this Coursework Specification in class. However, if you have any related questions, please feel free to contact the Module Leader on MS-Teams or via email at iatropoi@lsbu.ac.uk as soon as possible.

Screenshot Requirements Example

The commands to add a new user in a Linux machine are the following:

```
root@raspberrypi:~# useradd -s /bin/bash -m ioaniatr80
root@raspberrypi:~# usermod -aG sudo ioaniatr80
root@raspberrypi:~# passwd ioaniatr80
New password:
Retype new password:
passwd: password updated successfully
root@raspberrypi:~# su ioaniatr80
ioaniatr80@raspberrypi:~$ whoami
ioaniatr80
```

The example below, clearly shows the IP address of this machine by using 'ifconfig', and 'ssh' commands.



```
ioaniatr80@kali-2019: ~
ioaniatr80@kali-2019: ~ 102x26
ioaniatr80@kali-2019:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.10.10.6 netmask 255.255.255.0 broadcast 10.10.10.255
    inet6 fe80::a00:27ff:fe8f:4eca prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:8f:4e:ca txqueuelen 1000 (Ethernet)
    RX packets 129 bytes 19244 (18.7 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 661 bytes 50408 (49.2 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 20 bytes 1116 (1.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 20 bytes 1116 (1.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ioaniatr80@kali-2019:~$ ssh 10.10.10.100
The authenticity of host '10.10.10.100 (10.10.10.100)' can't be established.
ECDSA key fingerprint is SHA256:EWptTr0Xn9NMudUhcD3+AMXSigXAGS4uldZp3grLm8w.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '10.10.10.100' (ECDSA) to the list of known hosts.
ioaniatr80@10.10.10.100's password: █
```

Figure 1: Screenshot Example (ifconfig execution)

Resources

All the module's lectures, tutorial handouts, and references are recommended in the module guide. Some links that may be found useful:

Getting started with Raspberry-Pi: [Link](#)
Networking Foundations (LinkedIn Learning) : [Link](#)
Kali-Linux Revealed (Online free course): [Link](#)
Wireshark Filtering: [Link](#)
ZoneMinder on a Raspberry Pi: [Link](#)
Kodi on a Raspberry Pi (OSMC): [Link](#)
Installing LibreELEC on Raspberry Pi: [Link](#)
DOSBox Emulator: [Link](#)
FreeDOS: [Link](#)
Vetusware: [Link](#)
WinWoirdPC: [Link](#)
Top 30 Linux Monitoring Tools: [Link](#)
How to measure GPU: [Link](#)
GLMark2: [Link](#)
Geekbench: [Link](#)
PSUtil: [Link](#)
DOS Benchmark tools Pack: [Link](#)

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